Classical training of conditional instantaneous quantum polynomial-time Born machines

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Instantaneous quantum polynomial-time (IQP) Born machines admit classical estimation of training observables for suitable input states, allowing model optimization without quantum circuit sampling. However, their standard fixed-input formulation models a single distribution. In this work, we introduce a conditional framework that encodes conditions into input states before applying a learned IQP unitary shared across conditions. The input preparation need not have strict IQP form, provided its reference state supports efficient classical sampling and amplitude-ratio evaluation. These operations enable unbiased classical estimation of Pauli-Z moments and their gradients. Experiments on a sixteen-qubit discrete Gaussian family and the Curie–Weiss model demonstrate conditional generation. For Curie–Weiss, the shared model achieves lower mean full-distribution total variation distance than both independent baseline families at two of three held-out temperatures, although the baselines are fitted directly at those temperatures and each family uses three times the total parameters and optimizer updates. Additional runs demonstrate the computational feasibility of classical conditional training from 64 to 2048 qubits. Furthermore, we show how input-state encoding can impose exact output symmetries in both conditional and unconditional IQP Born machines.

I. INTRODUCTION

Conditional generative models approximate a family of target distributions $p(\cdot \mid c)$ with a model $q_{\boldsymbol{\theta}}(\cdot \mid c)$, where c denotes the condition and $\boldsymbol{\theta}$ contains learned parameters shared across conditions. Classical examples include conditional generative adversarial networks [1–3], conditional variational autoencoders [4–6], and conditional diffusion models [7–9]. In the quantum domain, researchers have also adapted quantum circuit Born machines (QCBMs) [10, 11] to conditional tasks [12]. Researchers can introduce conditioning through input-state preparation or dedicated condition registers before applying a shared parameterized circuit [13–16]. Alternatively, postselection on a learned joint Born distribution can also yield conditional samples [17]. Beyond QCBMs, quantum diffusion models have explored conditional generation of both classical data and quantum states [18–20].

Existing complexity-theoretic results support the conjecture that classical algorithms cannot efficiently sample from certain output families of instantaneous quantum polynomial-time (IQP) Born machines under standard assumptions [21–25]. Yet, for maximum mean discrepancy (MMD) training with a Gaussian kernel [26], as commonly used for QCBMs [10, 27], classical algorithms can efficiently estimate the required Pauli-Z expectation values and their parameter gradients [28]. These estimators build on Van den Nest's techniques for computationally tractable input states [29]. However, the standard IQP formulation models a single target distribution per circuit. Extending IQP Born machines to conditional generation requires a conditioning mechanism that preserves classical access to the training observables and their gradients.

In this work, we develop a classically trainable framework for conditional IQP Born machines. We apply one learned IQP unitary $U_{\boldsymbol{\theta}}$ to condition-dependent input states $|\phi(c)\rangle$, giving $q_{\boldsymbol{\theta}}(x \mid c) = |\langle x|U_{\boldsymbol{\theta}}|\phi(c)\rangle|^2$. We establish sufficient conditions on the encoder for efficient classical estimation of the conditional training observables and their gradients, and construct encoders satisfying these conditions. Specifically, for the reference amplitudes $\psi_c(z) = \langle z|H^{\otimes n}|\phi(c)\rangle$, we require efficient sampling from $|\psi_c|^2$ and evaluation of amplitude ratios.

Building on sparse-observable methods [29, 30] and IQP training methods [28], we derive unbiased estimators of conditional Pauli-Z moments and their derivatives with respect to the learned phases. We bound the second moments of the moment and derivative estimators by one and four, respectively. These bounds imply that $O(\epsilon^{-2})$ reference samples suffice to estimate each specified moment or derivative to additive error ϵ at fixed confidence. We combine these estimators to obtain unbiased finite-sample estimates of the conditional MMD^2 objective and its gradients, enabling classical optimization of a single IQP Born machine across conditions. Figure 1 summarizes the framework.

We evaluate the framework on a translated discrete Gaussian family and Curie–Weiss distributions at sixteen qubits. For the Gaussian family, we train a single conditional model at five distinct conditions. The model

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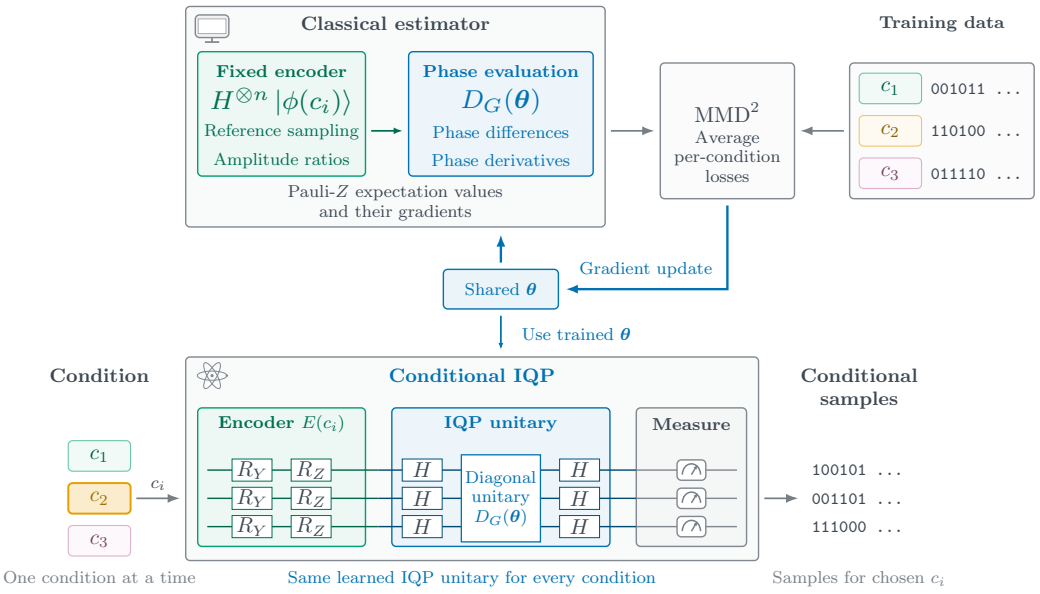


FIG. 1. Proposed framework for classical training and quantum sampling of conditional instantaneous quantum polynomial-time (IQP) Born machines. An encoder maps each condition c_i to an input state $|\phi(c_i)\rangle$. During classical training, samples and amplitude ratios from the reference state $H^{\otimes n}|\phi(c_i)\rangle$, together with phase differences induced by the IQP layer, yield estimates of conditional Pauli-Z moments and their gradients. These estimates enter the squared-MMD objective, averaged across training conditions to optimize one shared parameter vector $\boldsymbol{\theta}$. During quantum sampling, a chosen condition c_i determines the input state. Applying the trained IQP unitary $U_{\boldsymbol{\theta}}$ and measuring in the computational basis produces samples from the learned conditional distribution $q_{\boldsymbol{\theta}}(\cdot \mid c_i)$.

learns to translate the output profiles in response to the condition that we encode in the input state.

For the Curie–Weiss model, we train a single model at four inverse temperatures. The model captures the change from unimodal to bimodal magnetization profiles, including the emergence of bimodality at the held-out inverse temperature $\beta = 1.0$. The encoder represents temperature through input phases while maintaining uniform computational-basis probabilities. The learned circuit converts this phase dependence into temperature-dependent output distributions. We compare this model with conditional and unconditional baseline families. Each family contains three independent circuits that we fit directly at the evaluation temperatures and uses three times the total parameters and optimizer updates of the shared model. The shared model achieves lower mean full-distribution total variation distance (TVD) than both baseline families at two of the three held-out temperatures, including $\beta = 1.6$, beyond the training range.

We also assess computational feasibility through optimization runs from 64 to 2048 qubits, tracking sampled MMD^2 at held-out temperatures throughout training.

Input-state design also allows exact output symmetries without restricting the learned phases. We illustrate this capability with a fixed input that enforces symmetry under flipping a selected subset of qubits while preserving classical trainability. Appendix A 3 c extends the construction to temperature-dependent Curie–Weiss inputs and reports a matched encoder comparison.

The remainder of this paper is organized as follows. In Section II, we introduce the conditional IQP model and its classical estimators. In Section III, we develop the conditional MMD training protocol. In Section IV, we describe the numerical experiments and report their results. In Section V, we show how input-state preparation can enforce output symmetries. In Section VI, we discuss the implications and limitations of our findings, and in Section VII, we summarize the contributions.

II. CONDITIONAL IQP MODEL

We separate condition encoding from the learned unitary transformation. An encoder maps each condition $c \in \mathcal{C}$ to an input state $|\phi(c)\rangle$, and we apply the same parameterized IQP unitary $U_{\boldsymbol{\theta}}$ to every encoded state. For a target family $\{p(\cdot \mid c) : c \in \mathcal{C}\}$ over n-bit strings, we define the model distribution as

$$q_{\boldsymbol{\theta}}(x \mid c) = |\langle x| U_{\boldsymbol{\theta}} |\phi(c)\rangle|^2, \qquad x \in \{0,1\}^n. \tag{1}$$

Throughout this work, we use "IQP unitary" to denote only the learned block $U_{\boldsymbol{\theta}}$. The encoder need not have IQP form and may include non-diagonal operations, so the complete conditional circuit need not belong to the

strict IQP circuit class. Our conditions constrain classical access to the reference state rather than the encoder's gate structure: the reference state must support efficient sampling and amplitude-ratio evaluation.

The encoder also constrains how the model's output probabilities depend on the condition c. For rotation encodings with angles linear in c, the encoding generators determine the accessible Fourier frequencies, while the learned parameters control their coefficients within the constraints of the circuit architecture [31]. Multidimensional Fourier analyses further describe how the encoding and circuit architecture constrain the model's expressive capacity [32]. Caro et al. also derive encoding-dependent generalization bounds for parameterized quantum circuits [33]. We therefore distinguish encoder expressivity and generalization from the requirements for efficient classical estimation.

We parameterize the shared IQP unitary by an undirected graph $G = (V, E)$ with $n = |V|$ vertices. Each vertex $i \in V$ contributes a one-body generator Z_i, and each edge $\{i, j\} \in E$ contributes a two-body generator $Z_i Z_j$. We write the unitary as

$$U_{\boldsymbol{\theta}} = H^{\otimes n} D_G(\boldsymbol{\theta}) H^{\otimes n}, \tag{2}$$

where

$$D_G(\boldsymbol{\theta}) = \exp\left(i \sum_{i \in V} \theta_i Z_i + i \sum_{\{i,j\} \in E} \theta_{ij} Z_i Z_j \right). \tag{3}$$

All generators in D_G are diagonal in the computational basis and commute. In the numerical experiments, we use only one- and two-body generators. However, this formulation extends naturally to k-body interactions by adding the corresponding Pauli-Z products to the exponent.

We train the model using samples and amplitude ratios from the reference state $|\psi_c\rangle$ immediately before the diagonal IQP layer, without sampling the circuit's output distribution. We define its computational-basis amplitudes as

$$\psi_c(z) = \langle z | H^{\otimes n} | \phi(c) \rangle, \tag{4}$$

The squared amplitudes $|\psi_c(z)|^2$ give the reference sampling distribution for condition c. We require encoders that support two efficient classical operations for the relevant conditions c and bit masks a:

1. Sample $z \sim |\psi_c(z)|^2$.
2. Evaluate the amplitude ratio

$$R_c(z, a) = \frac{\psi_c(z \oplus a)}{\psi_c(z)} \tag{5}$$

for sampled z, where $\oplus$ denotes bitwise addition modulo two.

We call these two operations the sampling-and-ratio interface.

We estimate Pauli-Z moments by combining reference amplitude ratios with phase factors from the IQP layer. For each sampled string z, the ratio $R_c(z, a)$ compares the reference amplitudes of z and the paired string $z \oplus a$. The IQP layer contributes a complex phase factor determined by the phase difference between these strings. We multiply the amplitude ratio by this phase factor and average the real part over reference samples to estimate the corresponding Pauli-Z moment [29].

We evaluate amplitude ratios only on the support of $|\psi_c|^2$, because reference sampling never selects strings with $\psi_c(z) = 0$. Computationally tractable reference states in the sense of Van den Nest admit efficient computational-basis sampling and amplitude evaluation [29]. These states provide a sufficient realization of the sampling-and-ratio interface when amplitude evaluation supplies the precision needed to compute the required ratios efficiently.

The simplest case is a product-state encoder $|\phi(c)\rangle = \bigotimes_{i=1}^{n} |\phi_i(c)\rangle$. Its reference amplitudes factorize as $\psi_c(z) = \prod_i \psi_{c,i}(z_i)$, where $\psi_{c,i}(b) = \langle b | H | \phi_i(c) \rangle$. We can therefore generate reference samples by drawing each bit independently, and the amplitude ratio is

$$R_c(z, a) = \prod_{i : a_i = 1} \frac{\psi_{c,i}(1 - z_i)}{\psi_{c,i}(z_i)}. \tag{6}$$

When the local amplitudes have constant arithmetic evaluation cost, both operations require at most $O(n)$ local operations.

Efficient classical estimation of the observables used for training does not imply efficient classical sampling from $q_{\boldsymbol{\theta}}(\cdot \mid c)$. Existing IQP sampling-hardness results apply to specific circuit families and do not establish hardness for arbitrary condition-dependent input states or trained parameter values [22, 23]. To generate conditional samples on a quantum computer, we also need efficient circuits that prepare $|\phi(c)\rangle$ and implement $U_{\boldsymbol{\theta}}$. The classical sampling-and-ratio interface alone does not guarantee such circuits.

A. Conditional Pauli estimator and gradients

We express the conditional MMD objective in Section III in terms of Pauli-Z moments. For a bit mask $a \in \{0, 1\}^n$, we define $Z_a = \bigotimes_{i=1}^{n} Z_i^{a_i}$ and denote its conditional model expectation by $\mu_q(a \mid c)$. We use the sampling-and-ratio interface to derive estimators for this moment and its derivatives with respect to the learned phases.

We denote the diagonal IQP layer by $D_{\boldsymbol{\theta}}$ and write its action as

$$D_{\boldsymbol{\theta}} |z\rangle = e^{i f_{\boldsymbol{\theta}}(z)} |z\rangle, \tag{7}$$

with

$$f_{\boldsymbol{\theta}}(z) = \sum_{j=1}^{P} \theta_j \chi_{g_j}(z), \qquad \chi_{g_j}(z) = (-1)^{g_j \cdot z}, \tag{8}$$

where P is the number of generators and $g_j \in \{0,1\}^n$ specifies the support of the jth Pauli-Z generator. We evaluate the dot product $g_j \cdot z$ modulo two. We define the phase difference associated with the mask a as

$$\Delta_a f_{\boldsymbol{\theta}}(z) = f_{\boldsymbol{\theta}}(z \oplus a) - f_{\boldsymbol{\theta}}(z). \tag{9}$$

For a reference sample $z \sim |\psi_c|^2$, we define

$$\begin{aligned} W_{c,a}(z;\boldsymbol{\theta}) &= R_c(z,a) e^{i\Delta_a f_{\boldsymbol{\theta}}(z)}, \\ Y_{c,a}(z;\boldsymbol{\theta}) &= \mathrm{Re}\, W_{c,a}(z;\boldsymbol{\theta}). \end{aligned} \tag{10}$$

We define the derivative of the phase difference with respect to θ_j as

$$\begin{aligned} d_{j,a}(z) &= \partial_{\theta_j} \Delta_a f_{\boldsymbol{\theta}}(z) \\ &= \chi_{g_j}(z \oplus a) - \chi_{g_j}(z) \in \{-2, 0, 2\}. \end{aligned} \tag{11}$$

The following proposition establishes unbiased estimators for conditional Pauli-Z moments and their phase derivatives, and bounds their sampling error. We obtain these estimators by adapting amplitude-ratio methods for sparse observables [30]. For computationally tractable reference states, efficient estimation also follows from Van den Nest's sparse-operator theorem [29].

Proposition 1 (Classical estimation from reference-state access)**.** *Fix a condition c, a bit mask a, and a phase function with polynomially many diagonal Pauli-Z terms, as in Eq. (8). We assume that the reference state is normalized and independent of $\boldsymbol{\theta}$. We also assume that we can efficiently draw independent samples from $|\psi_c|^2$ and evaluate $R_c(z,a)$ for each sampled z. For a reference sample $z \sim |\psi_c|^2$, we define the moment and derivative estimators as follows:*

$$\begin{aligned} Y &= \mathrm{Re}\, W_{c,a}(z;\boldsymbol{\theta}), \\ Y'_j &= \mathrm{Re}[i\, d_{j,a}(z) W_{c,a}(z;\boldsymbol{\theta})] \end{aligned} \tag{12}$$

With exact sampling and arithmetic, these estimators have the following expectations and second-moment bounds:

$$\begin{aligned} \mathbb{E}[Y] &= \mu_q(a \mid c), & \mathbb{E}[Y^2] &\le 1, \\ \mathbb{E}[Y'_j] &= \partial_{\theta_j} \mu_q(a \mid c), & \mathbb{E}[(Y'_j)^2] &\le 4. \end{aligned} \tag{13}$$

We obtain unbiased estimates of the moment and its phase derivative by averaging Y and Y'_j over S independent reference samples. Their mean-square errors are at most $1/S$ and $4/S$, respectively. We can therefore estimate a specified moment or phase derivative to additive error ϵ at fixed confidence using $S = O(\epsilon^{-2})$ reference samples. These guarantees hold even when some reference amplitudes vanish.

Proof. We define the support of the reference distribution as $S_c = \{z : \psi_c(z) \neq 0\}$. Using $HZH = X$, we conjugate the measured observable Z_a through the final Hadamard layer to obtain $X_a = \bigotimes_{i=1}^n X_i^{a_i}$. This operator flips the bits selected by a, so

$$\begin{aligned} \mu_q(a \mid c) &= \langle \psi_c |\, D_{\boldsymbol{\theta}}^\dagger X_a D_{\boldsymbol{\theta}}\, | \psi_c \rangle \\ &= \mathrm{Re} \sum_{z \in S_c} \psi_c(z)^* \psi_c(z \oplus a) e^{i\Delta_a f_{\boldsymbol{\theta}}(z)} \\ &= \mathbb{E}_{z \sim |\psi_c|^2}[Y]. \end{aligned} \tag{14}$$

The last equality rewrites the sum as an expectation under $|\psi_c|^2$, using the amplitude ratio and the definition of Y.

The reference state does not depend on $\boldsymbol{\theta}$, so its sampling distribution and amplitude ratios remain fixed. We therefore differentiate only the IQP phase factor in the finite sum to obtain

$$\begin{aligned} \partial_{\theta_j} \mu_q(a \mid c) &= \mathbb{E}_{z \sim |\psi_c|^2}[\mathrm{Re}\,(i\, d_{j,a}(z) W_{c,a}(z;\boldsymbol{\theta}))] \\ &= \mathbb{E}[Y'_j]. \end{aligned} \tag{15}$$

Thus Y'_j is an unbiased estimator of the phase derivative.

Although individual amplitude ratios can be large, normalization bounds their second moment. The phase factor has unit magnitude, and the sampling probability $|\psi_c(z)|^2$ cancels the denominator in $|R_c(z,a)|^2$ when we take the expectation. We therefore obtain

$$\mathbb{E}[|W_{c,a}|^2] = \sum_{z \in S_c} |\psi_c(z \oplus a)|^2 \le 1. \tag{16}$$

The final inequality follows because bit flipping maps S_c to a subset of all bit strings and the reference state is normalized. Since Y is the real part of $W_{c,a}$, we have $\mathbb{E}[Y^2] \le 1$. Combining $|d_{j,a}(z)| \le 2$ with the same bound gives $\mathbb{E}[(Y'_j)^2] \le 4$. Averaging Y and Y'_j over S independent reference samples therefore gives mean-square errors at most $1/S$ and $4/S$, respectively. Chebyshev's inequality then yields the stated fixed-confidence sample complexity. The argument allows vanishing reference amplitudes because we evaluate ratios only for sampled strings. □

In practice, we draw S independent reference samples $z_s \sim |\psi_c|^2$ and estimate the moment as

$$\widehat{\mu}_q(a \mid c) = \frac{1}{S} \sum_{s=1}^{S} Y_{c,a}(z_s;\boldsymbol{\theta}). \tag{17}$$

We estimate the phase derivative by averaging the corresponding Y'_j values over the same samples. For the standard IQP input $|0\rangle^{\otimes n}$, all reference amplitudes are equal, so $R_c(z,a) = 1$. The moment estimator then reduces to $Y = \cos[\Delta_a f_{\boldsymbol{\theta}}(z)]$, recovering the usual IQP cosine estimator [29, 34].

For a circuit with r ancillary qubits, we apply the same estimator to the joint reference state, provided it supports the sampling-and-ratio interface. If we discard the

ancillary measurement outcomes, we extend each visible-bit mask a to $(a, 0^r)$, where 0^r denotes r zeros. The zero entries ensure that the observable acts as the identity on the ancillary qubits.

III. CONDITIONAL IQP TRAINING

We now formulate the training protocol for conditional IQP Born machines. Throughout this work, we use *classically trainable* to mean that we can efficiently estimate the training objective and its phase gradients to a prescribed additive accuracy on a classical computer.

A. Condition-balanced MMD objective

We optimize the shared parameter vector $\boldsymbol{\theta}$ by minimizing the squared maximum mean discrepancy (MMD) across the training conditions. For distributions p and q on $\{0,1\}^n$ and a positive-definite kernel k, the population squared MMD is [26]

$$\begin{aligned}\mathrm{MMD}^2(p,q) = \mathbb{E}_{x,x'\sim p}[k(x,x')] - 2\mathbb{E}_{x\sim p,\, y\sim q}[k(x,y)] \\ + \mathbb{E}_{y,y'\sim q}[k(y,y')],\end{aligned} \tag{18}$$

where we draw the two samples in each expectation independently.

Throughout this work, we use the Gaussian kernel on binary strings,

$$k_\sigma(x,y) = \exp\left[-\frac{d_H(x,y)}{2\sigma^2}\right], \tag{19}$$

where $d_H(x,y)$ is the Hamming distance, which counts the positions at which x and y differ, and $\sigma > 0$ is the kernel bandwidth.

Let $\mathcal{C}_{\mathrm{train}}$ denote the set of training conditions and τ the set of kernel bandwidths. We train the shared model by minimizing the condition-balanced objective

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{|\mathcal{C}_{\mathrm{train}}|}\sum_{c\in\mathcal{C}_{\mathrm{train}}}\frac{1}{|\tau|}\sum_{\sigma\in\tau}\mathrm{MMD}^2_\sigma(p(\cdot\mid c), q_{\boldsymbol{\theta}}(\cdot\mid c))\,. \tag{20}$$

Here, MMD^2_σ denotes the squared MMD computed with the kernel k_σ. We weight all training conditions and bandwidths equally. At each optimizer update, we independently draw one condition uniformly from $\mathcal{C}_{\mathrm{train}}$ and one bandwidth uniformly from τ, giving a stochastic estimate of this average.

B. Character expansion

For binary strings, the Gaussian kernel admits an exact expansion in binary characters. We sample a visible-bit mask $a \in \{0,1\}^n$ by drawing its coordinates independently according to

$$a_i \sim \mathrm{Bernoulli}(p_\sigma), \qquad p_\sigma = \frac{1-e^{-1/(2\sigma^2)}}{2}, \tag{21}$$

The corresponding binary character is

$$\chi_a(x) = (-1)^{a\cdot x}, \tag{22}$$

which records the parity of the bits selected by a. Averaging the product of these characters over a gives

$$k_\sigma(x,y) = \mathbb{E}_a[\chi_a(x)\chi_a(y)]\,. \tag{23}$$

For a condition c, define the target and model character moments

$$\begin{aligned}\mu_p(a\mid c) &= \mathbb{E}_{x\sim p(\cdot|c)}[\chi_a(x)],\\ \mu_q(a\mid c) &= \mathbb{E}_{x\sim q_{\boldsymbol{\theta}}(\cdot|c)}[\chi_a(x)].\end{aligned} \tag{24}$$

Substituting Eq. (23) into Eq. (18) gives

$$\mathrm{MMD}^2_\sigma(p(\cdot\mid c), q_{\boldsymbol{\theta}}(\cdot\mid c)) = \mathbb{E}_a\Big[(\mu_p(a\mid c) - \mu_q(a\mid c))^2\Big]\,. \tag{25}$$

Thus, the squared MMD averages the squared differences between target and model character moments over the mask distribution.

For a Born machine, the visible Pauli operator

$$Z_a = \bigotimes_{i=1}^n Z_i^{a_i} \tag{26}$$

has eigenvalue $\chi_a(x)$ on the computational-basis state $|x\rangle$. Its model expectation value is therefore

$$\mu_q(a\mid c) = \langle Z_a\rangle_{q_{\boldsymbol{\theta}}(\cdot|c)}. \tag{27}$$

We can thus express the conditional squared MMD entirely in terms of visible Pauli-Z moments. We estimate the model moments classically using the reference-state estimator in Sec. II A.

The bandwidth σ determines which correlation orders the objective emphasizes. The mask's Hamming weight $K = |a|$ counts the bits it selects, so it gives the correlation order of Z_a. Because we sample the mask coordinates independently,

$$\begin{aligned}K &\sim \mathrm{Binomial}(n, p_\sigma),\\ \mathbb{E}[K] &= np_\sigma,\\ \mathrm{Var}(K) &= np_\sigma(1-p_\sigma).\end{aligned} \tag{28}$$

Each bandwidth therefore induces a distribution over correlation orders. Large bandwidths favor lower-order correlations, whereas decreasing σ shifts weight toward higher-order correlations, with $\mathbb{E}[K] \to n/2$ as $\sigma \to 0$.

The choice of bandwidth can also influence the model's trainability. Previous results demonstrate that cost functions consisting of global terms can induce barren

plateaus [35, 36], even in shallow quantum circuits [37]. This creates a trade-off between trainability and sensitivity to higher-order correlations. Improving trainability may require restricting the objective to low-body correlations, which limits its ability to distinguish distributions that differ in higher-order correlations [38]. Researchers have proposed several strategies to address these difficulties, including tailored generator sets and restricted initializations [39, 40], and adversarial kernel adaptation [41]. How best to balance trainability and sensitivity to higher-order correlations remains an open question beyond the scope of this work.

C. Unbiased finite-sample MMD and gradients

For a fixed condition c and visible mask a, we draw $B \geq 2$ independent target samples and compute

$$T_b = \chi_a(x_b), \qquad x_b \sim p(\cdot \mid c). \tag{29}$$

Independently of these target samples, we draw $S \geq 2$ independent reference samples and evaluate the reference-state estimator from Eq. (10),

$$Y_s = Y_{c,a}(z_s; \boldsymbol{\theta}), \qquad z_s \sim |\psi_c|^2. \tag{30}$$

We denote the corresponding sample means by

$$\overline{T} = \frac{1}{B}\sum_{b=1}^{B} T_b, \qquad \overline{Y} = \frac{1}{S}\sum_{s=1}^{S} Y_s. \tag{31}$$

The direct estimate $(\overline{Y}-\overline{T})^2$ overestimates the squared moment difference on average because it includes the sampling variances of both sample means. We remove this bias by excluding products that use the same draw twice:

$$\begin{aligned}\widehat{\Delta^2_c}(a) = &\frac{1}{S(S-1)}\sum_{s\neq t} Y_s Y_t - 2\overline{Y}\,\overline{T} \\ &+ \frac{1}{B(B-1)}\sum_{b\neq b'} T_b T_{b'}.\end{aligned} \tag{32}$$

Independence within each pair gives

$$\mathbb{E}\left[\widehat{\Delta^2_c}(a)\right] = \left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)^2. \tag{33}$$

Thus, Eq. (32) provides an unbiased finite-sample estimator of the squared character-moment difference appearing in Eq. (25). An unbiased finite-sample estimate of MMD^2 may be negative.

We evaluate the pair sums using the identities

$$\sum_{s\neq t} Y_s Y_t = \left(\sum_s Y_s\right)^2 - \sum_s Y_s^2, \tag{34}$$

and, since $T_b^2 = 1$,

$$\sum_{b\neq b'} T_b T_{b'} = \left(\sum_b T_b\right)^2 - B. \tag{35}$$

Once we have computed the sample values, we can therefore evaluate the estimator in time linear in the numbers of target and reference samples.

The same construction gives an unbiased estimator of the phase gradients. For a phase parameter θ_j, let

$$Y'_s = \partial_{\theta_j} Y_{c,a}(z_s; \boldsymbol{\theta}). \tag{36}$$

We evaluate Y'_s using Eq. (15). Differentiating Eq. (32) gives

$$\partial_{\theta_j}\widehat{\Delta^2_c}(a) = \frac{2}{S(S-1)}\sum_{s\neq t} Y_s Y'_t - 2\overline{T}\,\overline{Y'}, \tag{37}$$

where

$$\overline{Y'} = \frac{1}{S}\sum_s Y'_s. \tag{38}$$

Taking the expectation yields

$$\mathbb{E}\left[\partial_{\theta_j}\widehat{\Delta^2_c}(a)\right] = 2\left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)\partial_{\theta_j}\mu_q(a \mid c), \tag{39}$$

which is the gradient of the corresponding squared moment difference.

At a fixed parameter vector, the expectations above average over independent draws from the target distribution and the reference sampling law. During training, we sample target minibatches with replacement from fixed training pools; the stochastic gradient is therefore unbiased for the empirical objective associated with those pools.

We exclude the diagonal terms $s = t$ because Y_s and Y'_s can be correlated when we compute them from the same reference draw. The fixed encoder makes the reference sampling law independent of $\boldsymbol{\theta}$. We therefore hold the sampled reference strings fixed during automatic differentiation to evaluate the phase gradients without differentiating through the sampler.

D. Stochastic mask estimator

Equation (25) expresses the squared MMD as an expectation over the mask distribution in Eq. (21). Since there are 2^n possible masks, we estimate this expectation using a finite sample of masks.

The identity mask $a = 0$ satisfies

$$\mu_p(0 \mid c) = \mu_q(0 \mid c) = 1 \tag{40}$$

for every condition and therefore contributes zero to the squared MMD. We consequently sample only nonidentity masks from the conditional distribution

$$P_\sigma(a \mid a \neq 0). \tag{41}$$

Under the original mask distribution, nonidentity masks have total probability

$$Q_\sigma^+ = 1 - (1 - p_\sigma)^n. \tag{42}$$

We multiply the conditional expectation by Q_σ^+ to preserve the original Gaussian kernel objective.

We draw M nonidentity masks $a_1, \ldots, a_M$ independently from $P_\sigma(a \mid a \neq 0)$ and estimate the squared MMD at condition c as

$$\widehat{\mathrm{MMD}}_\sigma^2(c) = Q_\sigma^+ \frac{1}{M} \sum_{m=1}^{M} \widehat{\Delta}_c^2(a_m), \tag{43}$$

where $\widehat{\Delta}_c^2$ is the unbiased finite-sample estimator in Eq. (32). Averaging over the mask, target, and reference draws recovers the full kernel squared MMD.

We evaluate each distinct sampled mask only once. Let $\mathcal{U}$ denote the set of distinct masks and m_u the number of times mask u occurs. We rewrite Eq. (43) as

$$\widehat{\mathrm{MMD}}_\sigma^2(c) = Q_\sigma^+ \sum_{u \in \mathcal{U}} \frac{m_u}{M} \widehat{\Delta}_c^2(u). \tag{44}$$

Weighting each distinct mask by m_u/M preserves the sampled objective and its gradient while reducing the number of Pauli moments we evaluate.

IV. NUMERICAL EXPERIMENTS

We evaluate the proposed architecture on a translated Gaussian family and the Curie–Weiss model at sixteen qubits. We also assess computational feasibility through optimization runs from 64 to 2048 qubits. For the sixteen-qubit benchmarks, we use the exact full-distribution total variation distance as the primary accuracy metric,

$$\mathrm{TVD}(p, q) = \frac{1}{2} \sum_{x \in \{0,1\}^n} |p(x) - q(x)|. \tag{45}$$

We also report the exact TVD between Hamming-weight profiles as a diagnostic,

$$\mathrm{TVD}_K(p, q) = \frac{1}{2} \sum_{k=0}^{n} |\rho_p(k) - \rho_q(k)|, \tag{46}$$

where $\rho_p(k)$ and $\rho_q(k)$ give the probabilities of observing Hamming weight k under p and q, respectively. Grouping bitstrings by Hamming weight gives $\mathrm{TVD}_K(p, q) \leq \mathrm{TVD}(p, q)$. The targets assign equal probability to all bitstrings with the same Hamming weight, but the learned models need not preserve this property. Profile TVD therefore does not resolve errors within Hamming-weight shells. We compute both metrics using the same exact circuit probabilities.

We train the models by minimizing the squared-MMD objective. We select Gaussian checkpoints using mean exact profile TVD over the development conditions. For all trained sixteen-qubit Curie–Weiss models, we use exact full-distribution TVD for hyperparameter and checkpoint selection. Appendix A specifies the training and evaluation protocols.

A. Discrete Gaussian translation

Our first experiment illustrates the conditioning mechanism with a translated discrete Gaussian Hamming-weight profile. For $n = 16$ bits, let $K = |x|$ denote the Hamming weight of bitstring x. The condition $c \in [0, 1]$ controls the nominal center $5 + 6c$ of the profile. The target distribution assigns equal probability to all bitstrings with the same Hamming weight.

We use a fixed product-state encoder that applies the same two rotations to every qubit:

$$|\phi(c)\rangle = \bigotimes_{i=1}^{16} R_Z\left(c - \frac{1}{2}\right) R_Y(1.5)\, |0\rangle_i \,. \tag{47}$$

Changing c modifies the input phases but leaves the encoder's computational-basis measurement distribution unchanged.

The shared unitary uses a 4×4 lattice with 16 one-body phase gates and 24 two-body phase gates between horizontal and vertical nearest neighbors, giving 40 trainable parameters.

We train at $c \in \{0, 0.25, 0.5, 0.75, 1\}$ and evaluate at 80 held-out conditions. The median across runs of the mean full-distribution TVD over these conditions is 0.0995, compared with 0.2922 for the encoder alone (Table I). The corresponding profile TVD is 0.0464. The higher full-distribution TVD reveals discrepancies within Hamming-weight shells that the profile metric does not resolve. Figure 2 shows the translated profiles and conditional response. Appendix A 2 gives the training and evaluation protocol.

TABLE I. Exact full-distribution and profile TVDs for the Gaussian benchmark. Each entry reports the median across five runs of the mean error over 80 held-out conditions. Bold entries mark the lowest error in each column.

Model	Full TVD ↓	Profile TVD ↓
Conditional IQP	**0.0995**	**0.0464**
Encoder only	0.2922	0.2922

B. Curie-Weiss spin model

We use the zero-field Curie–Weiss model [42] to test whether the architecture captures changes in distributional shape. The spins $s_i \in \{-1, +1\}$ interact with

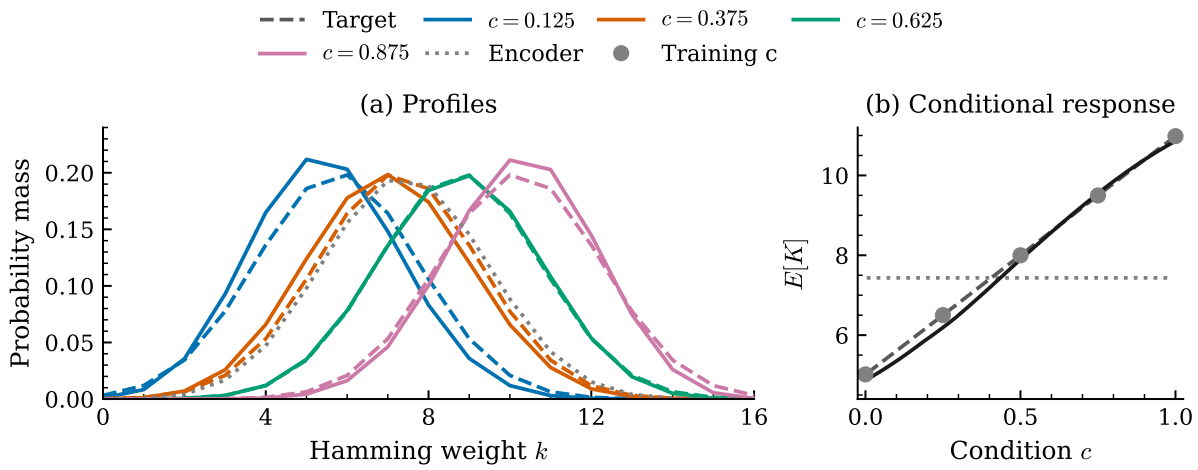


FIG. 2. Conditional translation of discrete Gaussian Hamming-weight profiles at sixteen qubits. (a) Solid colored curves show exact model profiles averaged across five runs at the development conditions $c \in \{0.125, 0.375, 0.625, 0.875\}$. We use these conditions for checkpoint selection and exclude them from fitting. (b) The solid black curve shows the median across runs of the model's mean output Hamming weight at each of 80 held-out conditions. Shading shows the interquartile range. Dashed curves show exact targets, dotted gray curves show the encoder alone, and gray circles mark training conditions on the target response. Table I reports exact full-distribution and profile errors.

equal strength across all pairs, so the target assigns equal probability to bitstrings with the same Hamming weight. As the inverse temperature β increases, the finite-size Hamming-weight profile changes from a central peak to two symmetric modes.

We encode the inverse temperature using the fixed product-state encoder

$$|\phi(\beta)\rangle = \bigotimes_{i=1}^{16} R_Z(0.75\beta) R_Y\left(\frac{\pi}{2}\right) |0\rangle_i \,. \tag{48}$$

The R_Z rotations encode β in the input phases while leaving the computational-basis measurement distribution uniform. We parameterize the IQP unitary with all one- and two-body generators, giving 136 trainable parameters. We train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$ and evaluate at the held-out inverse temperatures $\beta \in \{0.4, 1.0, 1.6\}$.

We compare the shared conditional IQP model with two independently tuned baseline families. Each family comprises three circuits, which we fit separately at the three evaluation temperatures. The independent conditional IQPs retain the corresponding temperature encoder, whereas the independent unconditional IQPs use $|0\rangle^{\otimes 16}$ as their input state.

We use exact full-distribution TVD for hyperparameter optimization (HPO) and model selection. We tune the shared model on development temperatures and each independent family jointly across its fitted temperatures. Appendix A 3 a details the protocol.

Each independent baseline family uses three circuits and therefore three times the total parameters and optimizer updates of the shared model. The shared model uses more training-pool samples, with 4,096 entries per seed compared with 3,072 for each independent family. Appendix A 3 reports the detailed budgets.

The shared conditional model captures the change from unimodal to bimodal magnetization profiles, including bimodality at the held-out inverse temperature $\beta = 1.0$ (Figure 3). It achieves lower mean full-distribution TVD than both independent baseline families at $\beta = 1.0$ and 1.6, with mean errors of 0.2175 and 0.1709, respectively (Table II). The temperature $\beta = 1.6$ lies beyond the training range. At $\beta = 0.4$, the independent conditional family achieves lower mean TVD than the shared model (0.1240 versus 0.2240), while the uniform encoder-only baseline achieves the lowest error (0.1180). Together with the held-out results, this comparison suggests that learning across conditions may provide a useful inductive bias for generalization.

The profile diagnostic in Appendix A 3 (Table VI) gives a different ranking from full-distribution TVD. At $\beta = 1.0$, both independent baseline families achieve lower mean profile TVD than the shared model but higher mean full-distribution TVD. At $\beta = 0.4$, the shared model reduces mean profile TVD relative to the encoder while increasing mean full-distribution TVD. These differences show that agreement in Hamming-weight profiles does not ensure accurate bitstring probabilities.

We also evaluate physical observables to distinguish magnetization magnitude from energy fluctuations. The exact zero-field target has $\langle m \rangle = 0$ at finite size, so we report the mean absolute magnetization $\langle |m| \rangle$, with $m = n^{-1} \sum_i s_i$ and $n = 16$. For the Hamiltonian $H = -nm^2/2$, we define the per-spin heat-capacity fluc-

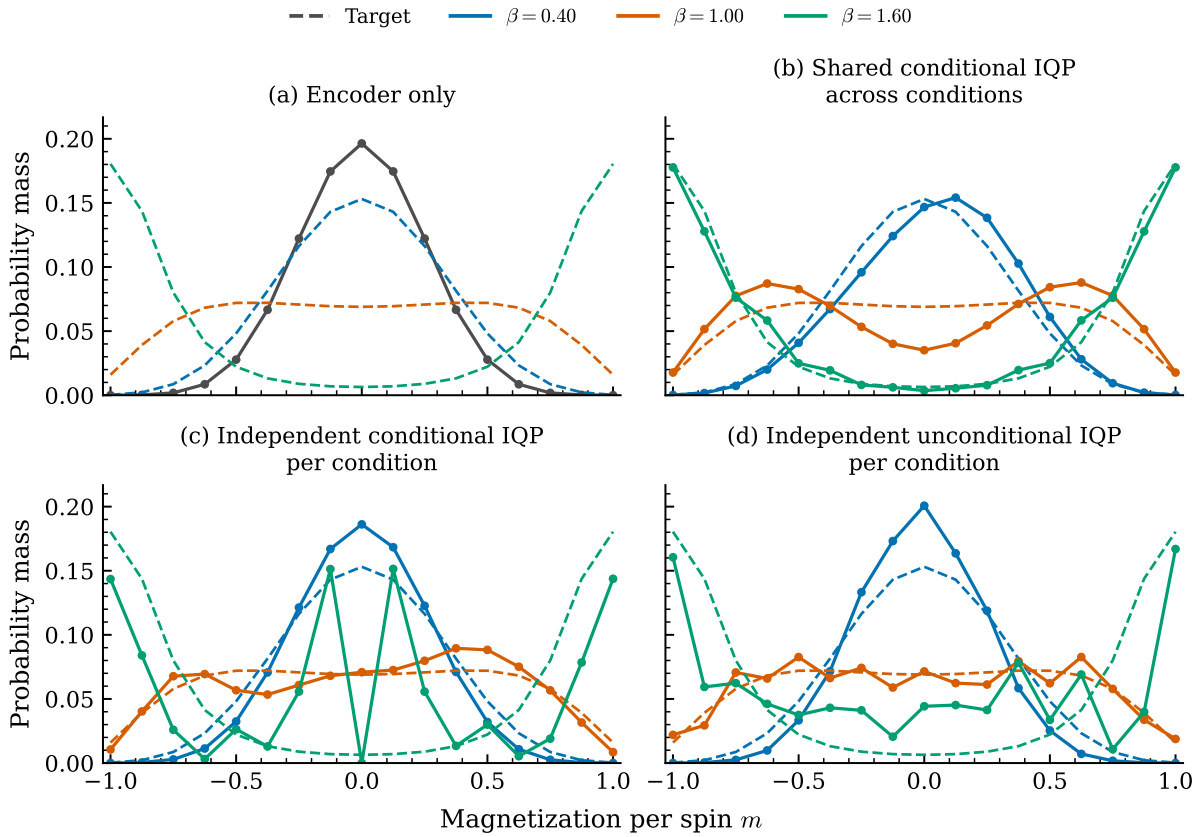


FIG. 3. Curie–Weiss magnetization profiles at sixteen qubits, with $m = (2k - 16)/16$. Panels show (a) encoder only, (b) shared conditional IQP, (c) independent conditional IQPs, and (d) independent unconditional IQPs. Colors distinguish $\beta = 0.4, 1.0, 1.6$, and dashed curves show exact targets. Solid curves show exact model profiles averaged over five confirmation runs used for model selection and reporting; the encoder-only profile is deterministic. We evaluate the shared model at temperatures excluded from fitting and current model selection, whereas we fit the independent circuits directly at each displayed temperature. Table II reports full-distribution TVD.

TABLE II. Exact full-distribution TVD for sixteen-qubit Curie–Weiss models. For trained models, we report means ± sample standard deviations across five confirmation runs used for model selection and reporting. The encoder-only baseline is deterministic. We evaluate the shared model at temperatures excluded from fitting and current model selection, whereas we fit the independent circuits directly at each evaluation temperature. Bold entries identify the lowest mean TVD in each column, including the encoder-only baseline.

Model	Evaluation	Full-distribution TVD ↓		
		$\beta = 0.4$	$\beta = 1.0$	$\beta = 1.6$
Encoder only	Baseline	**0.1180**	0.4406	0.8699
Shared conditional IQP	Held-out	0.2240 ± 0.0115	$\mathbf{0.2175 \pm 0.0136}$	$\mathbf{0.1709 \pm 0.0136}$
Independent conditional IQPs	Fitted	0.1240 ± 0.0083	0.2309 ± 0.0319	0.3948 ± 0.0003
Independent unconditional IQPs	Fitted	0.3842 ± 0.0017	0.3322 ± 0.0105	0.3299 ± 0.0119

tuation estimate

$$c_{\mathrm{fluc}}(\beta) = \frac{\beta^2}{n}\big(\langle H^2\rangle - \langle H\rangle^2\big). \tag{49}$$

This expression equals the equilibrium heat capacity for the Gibbs target. For a learned distribution, it measures energy fluctuations and need not equal the temperature derivative of its mean energy. We compute both observables from exact probabilities for the same five selected confirmation circuits on a grid from $\beta = 0$ to 2 in steps of 0.02, without additional fitting or model selection. The sweep includes fitting and development temperatures as well as temperatures beyond the fitting range; it is not an entirely held-out evaluation. Appendix A 3 b specifies the calculation and aggregation.

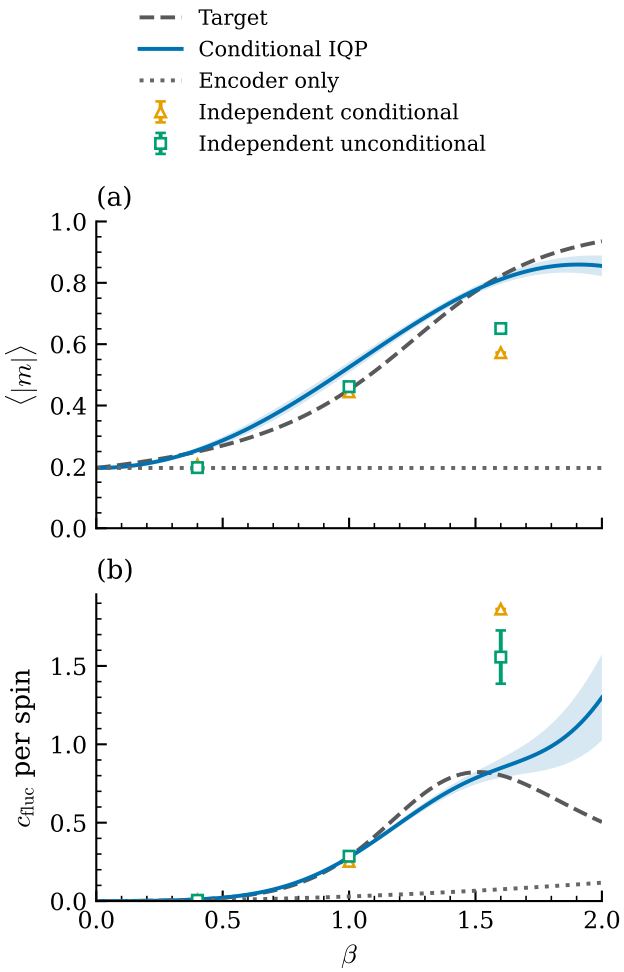


FIG. 4. Sixteen-qubit Curie–Weiss observables: (a) mean absolute magnetization and (b) per-spin heat-capacity fluctuation estimate, Eq. (49). Dashed and dotted curves show the exact target and uniform encoder reference. Conditional IQP curves and bands show mean and sample standard deviation across five frozen confirmation circuits at 101 temperatures. Independent markers use separately fitted circuits at $\beta = 0.4, 1.0, 1.6$, with the same summaries. We compute nonlinear observables within each run before aggregation. The sweep includes fitting, development and extrapolation conditions. For learned distributions, panel (b) measures energy fluctuations rather than a verified temperature response.

The dense response reveals errors that the three-temperature distribution comparison does not fully describe (Figure 4). At $\beta = 1.0$, the shared model overestimates mean absolute magnetization (0.525 versus the target 0.451), while the independent fitted baselines are closer. At higher inverse temperatures, the shared response levels off and underestimates the target (0.855 versus 0.936 at $\beta = 2$). Energy fluctuations show a different discrepancy. The target's heat capacity reaches its maximum on the evaluated grid near $\beta = 1.52$ and then decreases, whereas the shared model's fluctuation estimate continues increasing through $\beta = 2$, reaching 1.299 versus the target 0.504. At $\beta = 1.6$, however, the shared estimate remains closer to the target than either independent baseline. Thus, capturing the change in magnetization magnitude does not ensure recovery of the target's energy fluctuations, and the relative accuracy depends on the observable and temperature.

C. Scaling

We assess the computational feasibility of classical training at $n \in \{64, 128, 256, 512, 1024, 2048\}$ visible qubits, without ancillas. We use the fixed product-state encoder $R_Z(0.75\beta)R_Y(\pi/2)$ on every qubit. The learned unitary uses independent one-body phase gates and two-body phase gates between horizontal and vertical nearest neighbors on an open rectangular lattice. For an $L_x \times L_y$ lattice with $n = L_x L_y$, the model has $3n - L_x - L_y$ trainable parameters, so the parameter count grows linearly with n. Appendix A 4 gives the lattice dimensions and optimization settings.

At each system size, we train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$ and track the mean sampled MMD^2 over the held-out temperatures throughout optimization (Figure 5). Appendix A 4 specifies the training and reporting protocol.

The scaling study demonstrates the computational feasibility of classical conditional training up to 2048 qubits (Figure 5). At 64 qubits, the median held-out sampled MMD^2 decreases from 0.0504 to 0.0350 after 5,000 updates; at 2048 qubits, it decreases from 0.0490 to 0.0417. These modest reductions suggest limited progress under the chosen training configuration and may reflect suboptimal hyperparameters. The runs use a sparse lattice ansatz and optimizer settings that differ from the tuned sixteen-qubit comparison. Further tuning and benchmarking are needed for a systematic assessment of generative performance. Because the kernel bandwidths vary with system size, the absolute MMD^2 values do not directly compare distributional fidelity across sizes.

V. INPUT-STATE INDUCED SYMMETRIES IN IQP BORN MACHINES

Input-state encoding can also enforce exact output symmetries without restricting the learned phases. For a nonzero bit mask $s \in \{0,1\}^n$, the bit-flip operator $X_s = \bigotimes_{i=1}^n X_i^{s_i}$ commutes with the IQP unitary. If each encoded input is an eigenstate of X_s, the output state retains the same eigenvalue. Its computational-basis amplitudes at x and $x \oplus s$ therefore differ at most by a sign, giving equal probabilities for every condition and parameter vector. This applies the commuting-symmetry construction of Recio-Armengol *et al.* [28, Sec. 6] to conditional inputs. Such constraints provide inductive biases in variational quantum models [43].

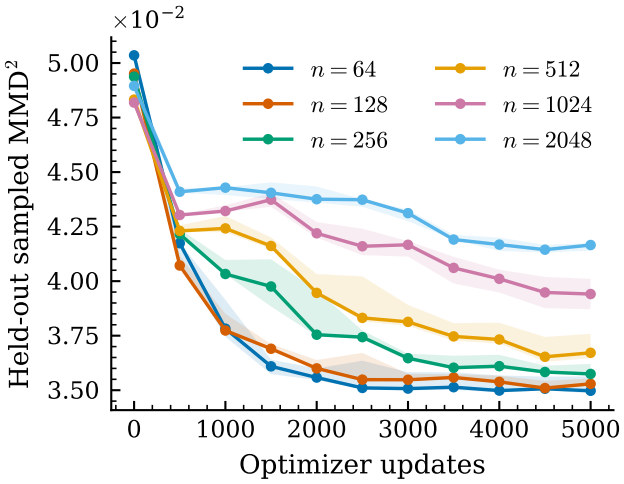


FIG. 5. Classical training of conditional IQP Born machines from 64 to 2048 qubits. At each plotted update, we average sampled MMD^2 over the held-out inverse temperatures $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$. Curves and shaded bands show the median and interquartile range of these averages across three independent training runs. We fix the expected mask weights at $w \in \{2, 4\}$ for each system size, so the corresponding kernel bandwidths vary with n.

A simple unconditional example uses the input

$$|\phi_s\rangle = \frac{|0^n\rangle + |s\rangle}{\sqrt{2}}. \tag{50}$$

This state satisfies $X_s |\phi_s\rangle = |\phi_s\rangle$ and therefore imposes

$$q_{\boldsymbol{\theta}}(x) = q_{\boldsymbol{\theta}}(x \oplus s) \qquad \text{for every } x \text{ and } \boldsymbol{\theta}. \tag{51}$$

The reference state $H^{\otimes n} |\phi_s\rangle$ has equal amplitudes on bitstrings satisfying $s \cdot z = 0$ modulo two and zero amplitudes elsewhere. To sample its measurement distribution, we choose a position with $s_i = 1$, draw the other bits uniformly, and set z_i to satisfy the parity constraint. For a sampled string z, the amplitude ratio $R_s(z, a)$ equals one when $s \cdot a = 0$ modulo two and zero otherwise. This input satisfies the sampling-and-ratio interface, so we can estimate the training moments and their gradients efficiently on a classical computer.

For three qubits, we choose $s = 110$, giving $|\phi_s\rangle = (|000\rangle + |110\rangle)/\sqrt{2}$. This input enforces $q_{\boldsymbol{\theta}}(x) = q_{\boldsymbol{\theta}}(x \oplus 110)$ for every parameter vector. The ordinary $|000\rangle$ input does not enforce this symmetry: with only $\theta_1 = \pi/6$ nonzero, the model assigns $q(000) = 3/4$ and $q(110) = 0$. Figure 6 compares the two inputs at a separate parameter setting, using the same unitary for both. Appendix A 3 c gives a temperature-dependent encoder that enforces global spin-flip invariance for Curie–Weiss and reports its effect on distributional accuracy.

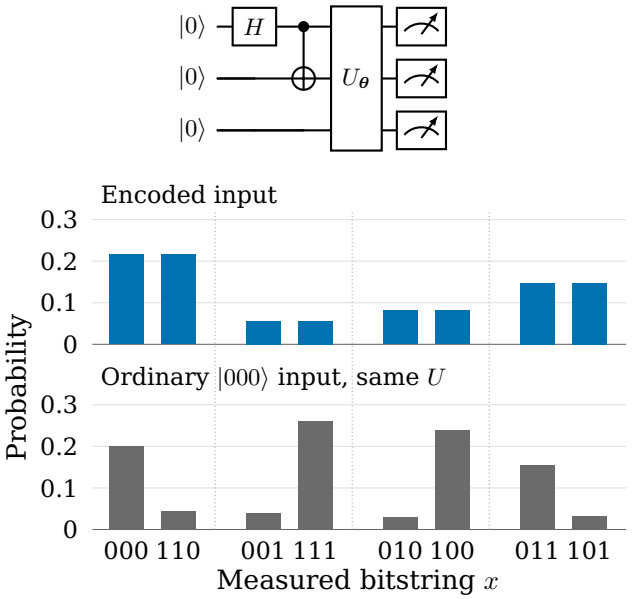


FIG. 6. The encoded input enforces equal probabilities for paired outcomes. The circuit prepares $(|000\rangle + |110\rangle)/\sqrt{2}$ before applying $U_{\boldsymbol{\theta}}$. The plots show exact probabilities at one parameter setting for the encoded input (blue) and the ordinary $|000\rangle$ input (gray), using the same unitary in both cases. Adjacent outcomes differ by the bit flip 110. The encoded input guarantees equal probabilities within each pair for every choice of learned phases.

VI. DISCUSSION

Our framework establishes sufficient conditions for classical training of conditional IQP Born machines with input preparation beyond strict IQP form. Efficient sampling and amplitude-ratio evaluation of the reference state enable unbiased estimation of conditional training moments and their gradients, while one learned unitary shares parameters across conditions. The sixteen-qubit experiments demonstrate conditional translations and changes in distributional shape through input phases. For Curie–Weiss, the shared model achieves lower mean full-distribution TVD than both independent baseline families at $\beta = 1.0$ and 1.6, with the latter extending beyond the training range. Each baseline family fits these temperatures directly and uses three times the aggregate parameters and optimizer updates, whereas the shared model uses more training-pool samples. At $\beta = 0.4$, the uniform encoder-only distribution achieves the lowest error. Joint training may provide useful regularization, but comparisons with matched data and optimization budgets are needed to assess this explanation.

Input-state design can impose exact output symmetries while preserving classical trainability. The unconditional bit-flip input and conditional Curie–Weiss projection enforce their symmetries for every parameter vector. The encoder comparison in Appendix A 3 c shows temperature-dependent changes in

full-distribution TVD, so exact symmetry does not guarantee improved accuracy. Spin-flip invariance also does not enforce uniformity within Hamming-weight shells. Classical output symmetrization is a relevant control because averaging a distribution with its bitwise complement cannot increase TVD to a spin-flip-invariant target. Coherent projection introduces interference and generally differs from this mixture. Comparisons that include symmetrization in fitting and selection would test whether projection improves accuracy beyond classical enforcement of the same symmetry.

The methodological limitations concern distributional evaluation and the scope of the scaling evidence. Efficient, unbiased MMD estimation does not ensure accurate recovery of every relevant correlation [44]. Although the Gaussian kernel includes all correlation orders, its bandwidth weights them unequally, and finite sampling and optimization may fail to resolve weakly weighted discrepancies. Agreement in Hamming-weight profiles can therefore coexist with errors in individual bitstring probabilities, making full-distribution diagnostics necessary whenever feasible. For Curie–Weiss, reporting reuses the confirmation runs used to select hyperparameters, so the reported spread does not independently assess model selection. The reporting temperatures are excluded from current fitting and selection but were examined in earlier experiments. Fresh seeds and previously unexamined conditions, evaluated with configurations fixed in advance, would strengthen the generalization evidence. The sparse-lattice runs establish computational feasibility up to 2048 qubits, with modest reductions in held-out sampled MMD^2. Their connectivity and optimizer settings differ from the sixteen-qubit comparison. Fixed expected mask weights emphasize low-order correlations as the system grows, while size-dependent bandwidths prevent direct comparisons of distributional accuracy across sizes.

Future work should examine richer tractable encoders and optimization strategies that improve conditional accuracy. Correlated inputs and nonlinear or multidimensional encodings should be assessed alongside state-preparation and estimation costs. Conditional extensions of spectral Born machines would require corresponding sampling, amplitude-ratio and gradient estimators [45]. Generator selection, initialization and adaptive kernels merit investigation [39–41], including whether gradients from different conditions reinforce or cancel one another. Quantum deployment should assess how noise and circuit partitioning affect the conditional response and imposed symmetries [46, 47]. Practical quantum sampling advantage remains open: our Gaussian and Curie–Weiss targets admit efficient classical sampling. Demonstrations would require comparisons with strong classical generative models and sampling-hardness or verification arguments appropriate to the trained conditional circuits.

VII. CONCLUSIONS

We introduced a conditional framework for IQP Born machines that combines a shared learned unitary with encoders that need not have IQP form. Efficient reference-state sampling and amplitude-ratio evaluation enable unbiased classical estimation of conditional training observables and their gradients, allowing optimization without quantum circuit sampling. Input-state design also lets us impose exact output symmetries independently of the learned parameters.

Our sixteen-qubit experiments demonstrate conditional translations and changes in distributional shape learned through input phases. For the Curie–Weiss model, the shared model achieves lower mean full-distribution TVD than both independently fitted baseline families at two held-out temperatures, including one beyond the training range, with fewer aggregate parameters and optimizer updates. Additional runs establish the computational feasibility of classical conditional training up to 2048 qubits. Together, the theoretical constructions and numerical results establish conditional generation and symmetry control within a framework that supports classical training. Further work must assess distributional accuracy at larger scales and practical quantum advantage.

DATA AVAILABILITY

The code and data supporting this study are available on GitHub [48].

ACKNOWLEDGMENTS

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Appendix A: Hyperparameters and evaluation protocols

1. Common training settings

We train all models using the unbiased estimator of the squared-MMD objective. We keep the encoders fixed and initialize the circuit phases independently as $\theta_j \sim \mathcal{N}(0, s_\theta^2)$. At each update, we sample one fitting condition and one kernel bandwidth uniformly, then draw a target minibatch with replacement. We draw fresh nonidentity masks and apply the probability correction for excluding the identity mask. Mask budgets count draws, including repeated masks. We draw reference samples independently of the target minibatch and reuse them across masks. Table III summarizes the settings.

For each initialization, we reset the optimizer and draw new circuit phases while reusing the target pools. Each training seed includes all prescribed initializations, and we select the checkpoint with the lowest selection score across them. Reported uncertainty summarizes variation across independent training seeds.

We train the Gaussian and scaling models with `Adam` [49], using cosine learning-rate decay to one tenth of the initial rate. For the Curie–Weiss models, we use the optimizer selected by HPO and keep the learning rate constant. We set $b_1 = 0.9$, and use $b_2 = 0.999$ and $\epsilon = 10^{-8}$ for the `Adam` variants. We set weight decay to zero.

TABLE III. Training settings and budgets per circuit. Mask and reference budgets give the number of draws per update. We select Curie–Weiss hyperparameters and checkpoints using full-distribution TVD. Table V gives the search space for the optimizer, learning rate and initialization width. We report selected learning rates and initialization widths to three significant figures and display the expected mask weight 7.99 as 8. For Curie–Weiss, the update and initialization budgets apply to confirmation and evaluation; screening uses the smaller budgets specified in Sec. A 3 a.

Setting	Gaussian	Shared product input	Shared spin-flip input	Independent conditional	Independent unconditional	Scaling
Updates per initialization	1,024	2,000	2,000	2,000	2,000	5,000
Initializations per seed	1	3	3	3	3	3
Evaluation seeds	5	5	5	5	5	3
Training targets per condition	512	1,024	1,024	1,024	1,024	10,000
Target minibatch	128	256	256	256	256	450
Training mask draws	1,024	2,048	2,048	2,048	2,048	512
Training reference draws	1,024	2,048	2,048	2,048	2,048	1,024
Expected mask weights	$\{1\}$	$\{2, 4, 8\}$	$\{2, 4, 8\}$	$\{2, 4, 8\}$	$\{2, 4, 8\}$	$\{2, 4\}$
Checkpoint interval	32	100	100	100	100	25
Optimizer	`Adam`	`Adam`	`Adam`	`AdamW ScheduleFree`	`Adam`	`Adam`
Learning rate	0.005	0.0310	0.0100	0.0603	0.00844	0.01
Learning-rate schedule	Cosine	Constant	Constant	Constant	Constant	Cosine
Initialization width s_θ	0.25	0.696	0.500	0.119	0.117	0.1
Coefficient b_1	0.9	0.9	0.9	0.9	0.9	0.9

For an expected mask weight w, we set the Gaussian bandwidth to

$$\sigma(n, w) = \left[-2 \ln\left(1 - \frac{2w}{n}\right)\right]^{-1/2}, \qquad 0 < w < n/2, \tag{A1}$$

where n denotes the number of visible qubits and w is the expected mask weight before excluding the identity mask. We weight the bandwidths equally in the objective. For the sixteen-qubit Curie–Weiss models, we use $w \in \{2, 4, 7.99\}$ and display 7.99 as 8. Using 7.99 keeps $w < n/2$ and the bandwidth strictly positive.

We report rotation angles in radians and compute sixteen-qubit full and profile TVDs from exact circuit probabilities. Sampled-MMD diagnostics do not enter these calculations.

2. Discrete Gaussian translation

We use 16 visible qubits on an open 4×4 lattice. The learned unitary contains 16 one-body phase gates and 24 two-body phase gates between horizontal and vertical nearest neighbors. We prepare the input state with the fixed

product-state encoder

$$|\phi_{\mathrm{G}}(c)\rangle = \bigotimes_{i=1}^{16} R_Z(c-1/2)R_Y(1.5)\,|0\rangle_i\,. \tag{A2}$$

We choose the near-equatorial angle 1.5 radians empirically to avoid an input state whose output distribution the learned unitary cannot change. At $c = 0.5$, replacing this angle with $\pi/2$ would prepare $|+\rangle^{\otimes 16}$, a simultaneous eigenstate of every X-type generator in the IQP unitary. The learned parameters would then contribute only a global phase, leaving a uniform output distribution with a fixed binomial Hamming-weight profile. Using 1.5 removes this exact invariance and allows the learned parameters to modify the output distribution at $c = 0.5$.

We use the target Hamming-weight profile $\rho(k \mid c) \propto \exp[-(k-(5+6c))^2/8]$, where $k = |x|$. Within each Hamming-weight shell, we assign equal probability to all bitstrings. We train at $c \in \{0, 0.25, 0.5, 0.75, 1\}$, using 512 target samples per condition and 2560 per seed.

We evaluate the model at initialization and every 32 updates through update 1024, selecting the checkpoint with the lowest mean exact profile TVD over the development conditions $c \in \{0.125, 0.375, 0.625, 0.875\}$. We then evaluate the selected model at 80 held-out conditions, $c_j = (j + 1/3)/80$ for $j = 0, \ldots, 79$. We draw 1000 target samples per evaluation condition for auxiliary sampled-MMD diagnostics.

We compute full-distribution and profile TVDs from exact circuit probabilities. For each seed, we average each error over the evaluation conditions and report the median of these averages across five independent seeds. Figure 2 averages exact profiles across the five selected circuits at the development conditions and shows the median and interquartile range of their mean output Hamming weights at each held-out condition.

3. Curie–Weiss models

We use the sixteen-qubit target $p(x \mid \beta) \propto \exp[\beta(2|x| - 16)^2/32]$. For all trained sixteen-qubit Curie–Weiss models, we parameterize the learned unitary with 16 one-body phase gates and 120 two-body phase gates, without ancillas. For the shared product-input model, we prepare $R_Z(0.75\beta)R_Y(\pi/2)\,|0\rangle$ on every qubit. For the shared invariant model, we use the normalized projection of this product input onto the $+1$ eigenspace of the global spin-flip operator, as described in Sec. A 3 c.

We train both shared models at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$, use $\beta \in \{0.5, 0.7, 1.3, 1.5\}$ for model selection, and report performance at $\beta \in \{0.4, 1.0, 1.6\}$. Each independent baseline family comprises three circuits that we fit directly at these reporting temperatures, using either the corresponding product input or $|0\rangle^{\otimes 16}$. For each baseline family, we select one optimizer, learning rate and initialization width jointly for its three circuits.

We train each circuit using 1024 target samples per fitting temperature and seed, minibatches of 256, and 2048 mask draws and 2048 reference draws per update. We store parameters and optimizer states in double precision and evaluate phase contractions in single precision. Table IV summarizes the aggregate fitting resources.

For auxiliary sampled-MMD diagnostics, we use 2048 validation targets per fitting temperature and 2048 independently drawn test targets per reporting temperature. Each diagnostic uses 2048 mask draws per condition and bandwidth, and 2048 reference samples per condition shared across bandwidths. We do not use these diagnostics for model selection.

TABLE IV. Curie–Weiss fitting budgets per seed for confirmation and evaluation. Training-target counts sum the pool entries over fitting temperatures and exclude repeated minibatch draws. Optimizer-update counts include all three initializations of every circuit. Each shared model fits one circuit, and each independent baseline family fits three.

Model	Trained circuits	Training targets	Optimizer updates
Shared product input	1	4,096	6,000
Shared spin-flip input	1	4,096	6,000
Independent conditional	3	3,072	18,000
Independent unconditional	3	3,072	18,000

a. *Hyperparameter optimization*

We run one Optuna study for each of the four model families, using a tree-structured Parzen estimator (TPE) [50]. Each study evaluates 60 configurations. We tune the optimizer choice among `Adam`, `AdamWScheduleFree` and

`SGDScheduleFree`, the constant learning rate, and the Gaussian initialization width [49, 51]. Table V gives the search space. We assign each family the same number of trials, although each independent-family trial fits three circuits per seed.

We use random search as a reference for the trial budget [52]. For 60 independent configurations drawn from a fixed sampling distribution, the probability of sampling at least one configuration in its best-performing 5% is $1 - 0.95^{60} \approx 0.954$. This calculation motivates the budget but does not establish equivalent coverage for adaptive TPE proposals.

TABLE V. Curie–Weiss hyperparameter search space. We treat optimizer choice as categorical and search learning rates and Gaussian initialization widths on a logarithmic scale.

Hyperparameter	Options or range
Optimizer	`Adam`, `AdamWScheduleFree`, `SGDScheduleFree`
Learning rate, `Adam` variants	$[3 \times 10^{-4}, 10^{-1}]$
Learning rate, `SGDScheduleFree`	$[10^{-3}, 1]$
Initialization width s_θ	$[0.01, 0.7]$ radians

We begin each study with 12 startup trials, including one reference configuration per optimizer. We set the reference learning rates to 0.01 for the `Adam` variants and 0.1 for `SGDScheduleFree`, with $s_\theta = 0.5$ for all reference configurations. We generate proposals in batches of two and update the sampler in proposal order. We do not prune trials.

We fix the optimizer coefficients as specified in the common training settings and use no warmup or gradient clipping. For schedule-free optimizers, we set the learning-rate weighting exponent to 2 and score checkpoints using the averaged parameters provided for evaluation.

For each screening trial, we train every circuit for 1000 updates on each of three seeds, using one initialization per seed. We compute exact TVD from the full distribution over the 2^{16} computational-basis states at initialization and every 100 updates. For each shared model, we select the checkpoint with the lowest mean TVD over the development temperatures. Within each independent baseline family, we select each circuit's checkpoint using TVD at its fitted temperature and average the three selected errors within each seed. We then take the median of the three seed scores as the trial score.

We confirm the three best distinct configurations from each study on five fresh seeds, training each circuit for 2000 updates from each of three initializations per seed. We select each circuit's checkpoint across its initializations using the same TVD criterion as in screening, then choose the configuration with the lowest median score across the five seeds.

We report performance using the five selected confirmation runs, without further fitting or checkpoint selection. Because these runs also inform hyperparameter selection, they do not provide an independent final evaluation. In the current protocol, we exclude the shared models' reporting temperatures from fitting and selection, although earlier experiments examined them. We fit the independent circuits directly at their reporting temperatures. For each temperature, we report the mean and sample standard deviation across the five seeds.

TABLE VI. Exact Hamming-profile TVD for sixteen-qubit Curie–Weiss models. We report means ± sample standard deviations across the five confirmation runs used for model selection and reporting. We select models using full-distribution TVD; encoder-only values are deterministic. We evaluate the shared model at temperatures excluded from fitting and current selection, and fit each independent circuit at its reporting temperature. Bold entries identify the lowest mean error at each temperature, including the encoder-only baseline.

Model	Evaluation	Hamming-profile TVD ↓		
		$\beta = 0.4$	$\beta = 1.0$	$\beta = 1.6$
Encoder only	Baseline	0.1180	0.4406	0.8699
Shared conditional IQP	Held-out	0.0976 ± 0.0269	0.1301 ± 0.0317	$\mathbf{0.0631 \pm 0.0135}$
Independent conditional IQPs	Fitted	$\mathbf{0.0939 \pm 0.0073}$	0.0885 ± 0.0116	0.3948 ± 0.0003
Independent unconditional IQPs	Fitted	0.1186 ± 0.0040	$\mathbf{0.0780 \pm 0.0320}$	0.3293 ± 0.0119

Table VI reports exact Hamming-profile errors for the primary Curie–Weiss comparison.

b. *Physical observables*

For a distribution q, let $\rho_q(k \mid \beta)$ denote its exact Hamming-weight profile and set $m_k = (2k-n)/n$ with $n = 16$. We compute its moments as $\langle m^r \rangle_q = \sum_{k=0}^{n} \rho_q(k \mid \beta) m_k^r$, and its mean absolute magnetization as $\sum_k \rho_q(k \mid \beta)|m_k|$. The target law corresponds to $H = -nm^2/2$ in units $J = k_B = 1$. Excluding self-interactions would add only a constant to H and would not change the energy variance. Equation (49) therefore becomes

$$c_{\mathrm{fluc},q}(\beta) = \frac{\beta^2 n}{4} \left(\langle m^4 \rangle_q - \langle m^2 \rangle_q^2 \right). \tag{A3}$$

The observables depend only on the shell profile and do not test uniformity within shells. For the Gibbs target, differentiating the mean energy with respect to $T = \beta^{-1}$ gives this fluctuation expression. We do not assume that the learned family satisfies the same derivative identity.

We evaluate the shared product-input circuits selected by full-distribution TVD on 101 equally spaced inverse temperatures $\beta_j = 0.02j$, $j = 0, \ldots, 100$, using their frozen parameters and ordered generators. The grid includes the fitting temperatures $\{0.6, 0.8, 1.2, 1.4\}$ and development temperatures $\{0.5, 0.7, 1.3, 1.5\}$, as well as extrapolation below and above the fitting range. We reuse the five winning confirmation seeds 2026100610–2026100614; these are not an independent assessment of model selection. The selected checkpoint profiles and full-distribution TVDs at $\beta \in \{0.4, 1.0, 1.6\}$ reproduce the existing reporting records to numerical precision. We compute each observable separately within each seed before taking its mean and sample standard deviation, including the nonlinear heat-capacity estimate. The independent baselines appear only at their fitted temperatures, with the same within-seed calculation and aggregation. The uniform encoder reference has the binomial shell law $\rho_0(k) = 2^{-n}\binom{n}{k}$. These diagnostics do not change fitting, hyperparameter selection, or checkpoint selection.

c. *Spin-flip invariant temperature encoding*

The zero-field Curie–Weiss target satisfies $p(x \mid \beta) = p(\bar{x} \mid \beta)$, where $\bar{x}$ denotes the bitwise complement of x. We enforce this global spin-flip symmetry through the input state while retaining the temperature encoding and all 136 trainable circuit phases. With $n = 16$, $t = 0.75\beta$, and $F = X^{\otimes n}$, we project the product encoder onto the $+1$ eigenspace of F and normalize the resulting state:

$$|\phi_+(\beta)\rangle = \frac{(I + F)\,|\phi(\beta)\rangle}{\sqrt{2[1 + \cos^n(t)]}}. \tag{A4}$$

The identity $\langle \phi(\beta)|F|\phi(\beta)\rangle = \cos^n(t)$ gives the normalization factor. Because the IQP unitary commutes with F, it preserves the $+1$ eigenspace. The output therefore satisfies $q_{\boldsymbol{\theta},+}(x \mid \beta) = q_{\boldsymbol{\theta},+}(\bar{x} \mid \beta)$ for every parameter vector. We apply the symmetry construction of Recio-Armengol *et al.* [28, Sec. 6] to this conditional encoder following the criterion in Section V. The learned circuit can still break permutation symmetry, so this constraint does not guarantee equal probabilities within Hamming-weight shells.

The projected encoder preserves efficient classical access to the reference state $H^{\otimes n}\,|\phi_+(\beta)\rangle$. For a bitstring z with Hamming weight $k = |z|$, its amplitude is

$$\psi_{+,\beta}(z) = \frac{\mathbf{1}_{k\ \mathrm{even}}[\cos(t/2)]^{n-k}[-i\sin(t/2)]^k}{\sqrt{[1 + \cos^n(t)]/2}}. \tag{A5}$$

The indicator equals one for even k and zero otherwise, so the reference state has support only on bitstrings of even Hamming weight.

We draw k from a binomial distribution with n trials and success probability $\sin^2(t/2)$, conditioned on even k. We then choose a bitstring uniformly among those with Hamming weight k. For a sampled string z, the amplitude ratio is

$$R_{+,\beta}(z, a) = \begin{cases} [-i\tan(t/2)]^{|z \oplus a| - |z|}, & |a| \text{ even}, \\ 0, & |a| \text{ odd}. \end{cases} \tag{A6}$$

A mask with odd Hamming weight moves z outside the reference-state support, giving a zero amplitude ratio.

These sampling and amplitude-ratio calculations require polynomial resources, so the projected encoder supports the conditional estimators and MMD training objective developed above.

We can prepare the invariant input deterministically by tracking parity on the last qubit of the existing register. We initialize all n qubits in $|0\rangle$, set $t = 0.75\beta$, and write $Z_\ell^{(p)} = [1+(-1)^p \cos^\ell(t)]/2$ for the probability that ℓ independent reference bits have parity $p \in \{0,1\}$. At each step, p denotes the parity of the prepared prefix, and ℓ counts the remaining qubits, including the parity qubit. We apply a parity-controlled R_Y rotation to the next unprepared qubit, with angle

$$\alpha_p = 2\,\mathrm{atan2}\left(\sin(t/2)\sqrt{Z_{\ell-1}^{(p\oplus 1)}}, \cos(t/2)\sqrt{Z_{\ell-1}^{(p)}}\right). \tag{A7}$$

We then apply a controlled-NOT (CNOT) gate from the newly prepared qubit to the last qubit to update the parity. Repeating this procedure on the first $n-1$ qubits prepares the reference-state magnitudes conditioned on even parity. We apply a single-qubit phase gate that multiplies $|1\rangle$ by $-i$ on every qubit to obtain Eq. (A5), then apply Hadamards to all qubits to prepare Eq. (A4). The construction uses $O(n)$ gates without ancillas or postselection.

Because the IQP unitary commutes with the global spin-flip operator, we obtain

$$q_{\boldsymbol{\theta},+}(x \mid \beta) = \frac{|A_{\boldsymbol{\theta},\beta}(x) + A_{\boldsymbol{\theta},\beta}(\bar{x})|^2}{2[1+\cos^n(t)]}, \tag{A8}$$

where $A_{\boldsymbol{\theta},\beta}(x) = \langle x|U_{\boldsymbol{\theta}}|\phi(\beta)\rangle$. Interference between these amplitudes changes the model family, but does not guarantee a better fit.

We obtain the encoder-only distribution by setting the learned unitary to the identity: $q_{0,+}(x \mid \beta) = [1+\cos((2|x|-n)t)]/[2^n(1+\cos^n(t))]$. Because the Curie–Weiss target and both encoder-only distributions are uniform within each Hamming-weight shell, each encoder's full-distribution TVD equals its profile TVD.

We compare the product-state and spin-flip invariant encoders using separate HPO studies with matched budgets and the same fitting and development temperatures. We use exact full-distribution TVD for model selection and reuse the selected confirmation runs for reporting (Table VII). Section A 3 a gives the HPO protocol.

We compare the shared product-input and spin-flip-invariant models using the same HPO protocol and paired seeds. Within each seed, we reuse the same target pools and Gaussian initialization directions and follow the same condition, minibatch, bandwidth and mask schedules. The tuned initialization widths may differ. For reference sampling, we reuse random numbers but sample from the distribution associated with each input. We retain odd-weight masks in the objective, although global spin-flip symmetry makes the corresponding moments of the projected model vanish.

We quantify deviations from global spin-flip symmetry using

$$D_{\mathrm{flip}}(q \mid \beta) = \frac{1}{2}\sum_x |q(x \mid \beta) - q(\bar{x} \mid \beta)|. \tag{A9}$$

The diagnostic vanishes exactly for spin-flip invariant distributions. For the trained product model, we obtain mean D_{flip} values of 0.4354, 0.1092, and 0.0087 at $\beta = 0.4, 1.0, 1.6$, respectively. The projected input enforces $D_{\mathrm{flip}} = 0$ at every temperature, independently of optimization. Numerical evaluation confirms this guarantee to machine precision.

TABLE VII. Exact full-distribution TVD for the sixteen-qubit Curie–Weiss encoder comparison. We report means ± sample standard deviations across five paired confirmation runs used for model selection and reporting. Bold entries mark the lowest mean TVD at each temperature.

Model	$\beta = 0.4$	$\beta = 1.0$	$\beta = 1.6$
Product-state input	0.2240 ± 0.0115	0.2175 ± 0.0136	$\mathbf{0.1709 \pm 0.0136}$
Spin-flip invariant input	$\mathbf{0.0902 \pm 0.0184}$	$\mathbf{0.1889 \pm 0.0134}$	0.1805 ± 0.0276

The spin-flip invariant input achieves lower mean full-distribution TVD than the product input at $\beta = 0.4$ and 1.0 (Table VII). This improvement holds across all five paired runs at both temperatures. At $\beta = 1.6$, the product input achieves lower mean TVD, although the invariant input reduces TVD in two of the five paired runs.

Without the learned circuit, the product-state encoder has full-distribution TVDs of 0.1180, 0.4406, and 0.8699 at $\beta = 0.4, 1.0, 1.6$, respectively; the spin-flip invariant encoder gives 0.2561, 0.5536, and 0.9002. Input projection alone therefore does not improve the fit at these temperatures.

Table VIII summarizes paired full-distribution TVD differences. We compute these differences within each seed before calculating their mean and sample standard deviation. The spin-flip-invariant model achieves lower full-distribution TVD than the product-input model in all five paired runs at $\beta = 0.4$ and 1.0, and in two of five at $\beta = 1.6$.

TABLE VIII. Paired differences in exact full-distribution TVD for sixteen-qubit Curie–Weiss models. Within each seed, we subtract the product-input model's TVD from the spin-flip-invariant model's TVD. Negative values favor the invariant input. We report the mean and sample standard deviation (SD) of these differences across the five paired confirmation runs used for model selection and reporting.

Statistic	$\beta = 0.4$	$\beta = 1.0$	$\beta = 1.6$
Mean	−0.1338	−0.0287	0.0096
Sample SD	0.0199	0.0154	0.0218

4. Scaling

We use the product-state Curie–Weiss encoder on every qubit, with no ancillas, and adopt the open lattice architectures listed in Table IX. At size n, the target distribution is $p(x \mid \beta) \propto \exp[\beta(2|x| - n)^2/(2n)]$.

We train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$, using 10,000 target samples per condition and 40,000 per seed. We evaluate at $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$ and use three independent seeds at each size. We use the cosine-decay `Adam` settings in Table III, with 5000 updates per initialization and three initializations per seed. These settings differ from the selected sixteen-qubit configurations. We have not tested the transfer of those configurations or retuned the scaling models.

TABLE IX. Scaling architectures, including single-qubit and nearest-neighbor pair phases.

Visible qubits	Lattice	Phases
64	8×8	176
128	8×16	360
256	16×16	736
512	16×32	1,488
1,024	32×32	3,008
2,048	32×64	6,048

Checkpoint selection minimizes mean sampled validation MMD^2 across training temperatures and bandwidths, checking initialization and every 25 updates across three initializations. Held-out temperatures do not enter fitting or selection. We report the selected initialization at its initial checkpoint and every 500 updates through update 5,000, averaging five independent MMD estimates per seed and condition grid. Each estimate uses 2,000 target draws per condition, 512 masks per condition and bandwidth, and 1,024 reference draws per condition. We reuse the reporting sample sets across iterations and share reference samples across bandwidths. The plot shows the median and interquartile range across three seed averages. We retain negative unbiased MMD^2 estimates.